

INDERPREET KAUR

AI / ML | Full-Stack Development | Data Engineering

Surrey, BC | Canadian Citizen (TN Visa Eligible) | (778) 325-4020 | inderpreetkaur762@gmail.com

[Portfolio](#) | [LinkedIn](#) | <https://github.com/inderpreet-k>

PROFESSIONAL SUMMARY

BTech IT student, Dean's Honour Roll ×3. Built and deployed Sentinel AI (ML firewall, 92% accuracy, 0 critical false negatives) and Spending Spotlight (live React/OpenAI app). Manages \$800K–\$1.2M/month in financial documentation across a \$15–\$20M portfolio. Canadian Citizen, TN visa eligible, no H-1B required.

TECHNICAL SKILLS

- **Languages:** Python (Pandas, Scikit-Learn, NLTK), JavaScript (React), PHP, SQL, Java, HTML/CSS
- **AI / ML:** Random Forest Classifier, TF-IDF Vectorization (char-level n-grams), OpenAI API, Prompt Engineering, Scikit-Learn
- **Data & DB:** MySQL, SQLite, CSV-to-DB Pipelines, ERD & Relational Design, Data Validation, Jupyter, Google Colab
- **Frameworks:** Flask, React, XAMPP / Apache
- **Tools:** Git, Vercel, QuickBooks, WordPress
- **Concepts:** Agile/Scrum, SDLC, SQA, Black-Box Security Testing, Information Retrieval, REST APIs

EDUCATION

Bachelor of Technology, Information Technology

Expected 2027

Kwantlen Polytechnic University (KPU), Surrey, BC | GPA: 3.37/4.33 | Dean's Honour Roll: Fall 2024, Summer 2025, Fall 2025

- Relevant Coursework: Data Warehousing & Mining (A), Advanced Web App Development (A+), Search Engine Principles (A–), Software Quality Assurance, Object-Oriented Software Engineering (Agile), Project Management (A–), Statistics, Database Systems

Diploma, Computer Information Systems

2021

Kwantlen Polytechnic University (KPU), Surrey, BC

PROJECTS

Sentinel AI: Autonomous ML Security Firewall | *Python* | *Scikit-Learn* | *PHP* | *MySQL* | *JUnit 5*

2026

[GitHub](#)

- Achieved 92% overall test accuracy (23/25 cases) across 3 attack categories, SQL injection, XSS, and OS command injection, with 0 false negatives on all 5 Critical-severity test cases.
- Trained a Random Forest Classifier on 24,161 labeled payloads using character-level TF-IDF n-gram vectorization (n=1–4, vocab: 80K features); confidence threshold of 0.75 minimized false positives on natural language inputs.
- Detected mixed-case SQL obfuscation (e.g. sElEcT * fRoM users) at confidence 1.00, demonstrating that character-level n-gram analysis generalizes to evasion techniques that defeat word-level models.
- Engineered a Python autonomous response engine that monitors logs in real-time and executes IP-based blacklisting in MySQL with zero human intervention across a 50+ page PHP MVC application.
- Built a JUnit 5 cross-language integration test harness, 25 parameterized test cases covering known attack vectors, legitimate inputs, and natural language edge cases, end-to-end black-box validation from HTTP payload to MySQL blacklist decision.

Spending Spotlight: AI Bank Statement Analyzer | *React | JavaScript | OpenAI API | Vercel* 2025

[Live Demo](#)

- Built and deployed a React web app that accepts PDF bank/credit card statements, classifies transactions via the OpenAI API, and flags unexpected spending against user-defined categories.
- Diagnosed and resolved a real-world data quality problem: Chase statements injected names, addresses, and page headers as false transactions; implemented pattern filtering and client-side error handling to eliminate noise and improve classification accuracy.
- Engineered multi-step UI with category toggles, custom category creation/removal, drag-and-drop PDF upload with type/size validation (16MB limit), and an expected vs. unexpected spending breakdown view.

AI Text-to-SQL Query Generator | *Python | SQLite | OpenAI API | Jupyter* 2025

[Open Notebook](#)

- Prototype that loads a CSV dataset into SQLite, accepts natural-language questions, and generates + executes SQL SELECT queries via GPT, enabling non-technical users to query relational data without SQL.
- Engineered strict prompt constraints (temperature=0, system-message scoping, regex post-processing) to ensure only valid SELECT statements execute, preventing destructive query injection.

Recipe Search Engine, Multi-Factor Ranking | *Python | NLTK | Flask* 2024

- Contributed to an IR system on a 13,493-document corpus (10,832 unique terms, 1.28M postings) implementing an inverted index, Boolean retrieval, TF-IDF, and Cosine Similarity scoring.
- Ran preprocessing pipelines (tokenization, stopword removal, Porter stemming), evaluated Precision@10 ranking accuracy, and supported Flask web UI integration.

PROFESSIONAL EXPERIENCE

Office Administrator / Bookkeeper Nexgen Electrical Ltd., White Rock, BC Mar 2025 – Present

- Process approximately \$800K–\$1.2M/month in invoices, SOVs, POs, and work orders across a \$15–\$20M active project portfolio spanning 8+ concurrent electrical construction projects.
- Manage monthly payroll of \$100K–\$250K for 30+ employees using QuickBooks; maintain high-accuracy records and ensure timely, compliant disbursements.
- Coordinate inspection scheduling, compliance documentation, and field-to-office communication across multiple concurrent projects.

Office Assistant / Bookkeeper Element Immigration Inc., White Rock, BC May 2023 – Mar 2025

- Maintained CRM and database records for 50–150 active client files, managed payroll, bookkeeping, and financial reconciliations.
- Maintained company website and published content via WordPress and Wix.

Bookkeeper Enfield Solutions Inc., White Rock, BC Aug 2021 – May 2023

- Handled full-cycle bookkeeping: payroll processing, annual GST filings, corporate filings, and financial reporting support for a 5-person firm.